TC Cure Engine OussamaLabs

CURE RECORD
TOURING 205/55 R16 · demo serial D-24117
Tread TR-TOUR · 2026-09-29

TEST IDENTITY

TYRE: TOURING 205/55 R16 · demo serial D-24117	TREAD COMPOUND: TR-TOUR	SCAN INTERVAL: 5s	CHANNELS: 22 of 22
CURE START: 240s Manual input (240s)	PRESS OPEN: 840s Manual input (600s)	DEFLATION: 850s Manual input (610s)	EA SOURCE: Compound DNA v1
ANALYSED BY: demo.engineer R&D admin	TAT FLOOR (PRIMARY): ECM ≥ t90@150×1		

CURE VERDICT — WINDOW-AWARE

FAIL **2 position(s) require action, 2 marginal.**
Window-aware: **18 PASS** · **2 MARGINAL** · **2 FAIL. 2 under-cured.**

ENGINEERING — ROOT-CAUSE BY ZONE

AP-01 — HARDWARE (asymmetry). Inadequate cure coincides with ΔT 16.9°C on AIT/AIB, A/BT/A/BB (whole zone). Check steam-line / bladder / platen on the cold side before the recipe.

BC-01 — WITHIN TOLERANCE (ran hot). 2 position(s) over the fixed cap but inside BC-01's reversion window (highest reversion index 23% of the 10 %-loss point). No cure action; if recurring, review setpoint margin.

⚐ BC-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

IL-01 — NO ACTION. All positions cured within spec.

RC-01 — NO ACTION. All positions cured within spec.

⚐ RC-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

SW-01 — NO ACTION. All positions cured within spec.

⚐ SW-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

TB-01 — NO ACTION. All positions cured within spec.

⚐ TB-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

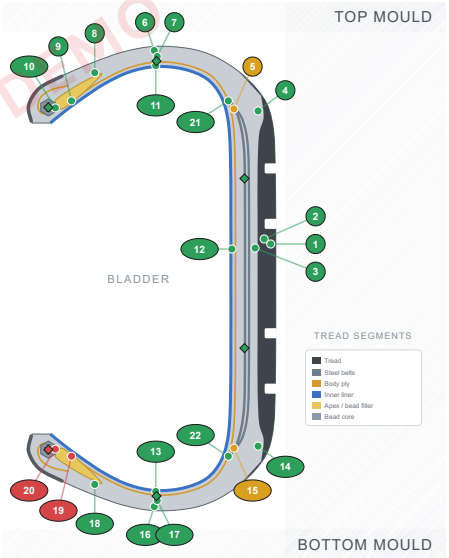
TR-TOUR — NO ACTION. All positions cured within spec.

⚐ TR-TOUR TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

UB-01 — NO ACTION. All positions cured within spec.

⚐ UB-01 TaT/ECM: Isothermal duration short, ramp contribution closes the gap — acceptable, no action.

SECTION MAP



PASS · MARGINAL · FAIL · excluded (fault)
not tested · estimated layer (no thermocouple)

DEMO · SYNTHETIC DATA · NOT FOR PRODUCTION USE

PRESS HEALTH — TOP/BOTTOM Δ @ DEFLATION (ADVISORY)

ZONE	COMPOUND	TOP°C	BOT°C	ΔT	STATUS
Tread Shoulder	TR-TOUR	171.1	170.8	0.3	PASS
Belt Edge	BC-01	172.1	171.7	0.4	PASS
Sidewall Upper	SW-01	170.9	167.4	3.5	WARNING
Sidewall Middle	SW-01	172.5	167.6	4.8	WARNING
Mud Stopper	RC-01	169	161.5	7.4	WARN-HIGH
Apex Inside	AP-01	170.6	155.7	14.9	WARN-HIGH
Apex/Bead	AP-01	168.3	151.4	16.9	WARN-HIGH
Inner Liner	IL-01	173.1	165.6	7.5	WARN-HIGH
Under Belt	UB-01	174.5	174.3	0.3	PASS

CEQ DETAILED RESULTS

CH	POSITION	TMAX	CEQ@PO	CEQ@DEF	%CEF	CEQ	ASSESS
1	TCT TR-TOUR	172.6	17.86	18.49	138.1	PASS	PASS
2	TCM TR-TOUR	173.3	15.66	16.32	121.1	PASS	PASS
3	TCB TB-01	174.6	15.7	16.42	155.1	PASS	PASS
4	TST TR-TOUR	171.4	12.65	13.24	97.9	PASS	PASS
5	SB2T BC-01	172.6	12.21	12.94	83.9	FAIL	MARGINAL
6	SWTMT SW-01	171.1	21.79	22.41	156.6	PASS	PASS
7	SWUMT SW-01	172.6	21.77	22.45	156.5	PASS	PASS
8	SWULT RC-01	169	13.63	14.19	110	PASS	PASS
9	AIT AP-01	170.7	12.43	13.05	116.1	PASS	PASS
10	A/BT AP-01	168.7	9.48	10.01	88.5	PASS	PASS
11	ILT IL-01	176.6	33.66	34.54	122	PASS	PASS
12	ILM IL-01	180.5	34.04	35.09	123.3	PASS	PASS
13	ILB IL-01	168.2	18.06	18.58	65.4	PASS	PASS
14	TSB TR-TOUR	171.2	12.45	13.03	96.3	PASS	PASS
15	SB2B BC-01	172.5	12	12.72	82.5	FAIL	MARGINAL
16	SWTMB SW-01	167.6	17.23	17.73	123.9	PASS	PASS
17	SWUMB SW-01	167.6	15.71	16.22	113	PASS	PASS
18	SWULB RC-01	161.8	8.42	8.77	67.9	PASS	PASS
19	AIB AP-01	156	4.38	4.62	40.9	FAIL	FAIL
20	A/BB AP-01	152.2	2.92	3.1	27.3	FAIL	FAIL
21	UBT UB-01	174.6	13.34	14.07	103	PASS	PASS
22	UBB UB-01	174.3	12.99	13.7	100.3	PASS	PASS

TAT — ECM ENVELOPE PER COMPOUND (PRIMARY) · RAW MIN≥150 SECONDARY

COMPOUND	ECM WORST	FLOOR	MARGIN	ECM MAX	TIER	RAW150 W/FLR	ENVELOPE
TR-TOUR MARCHING	27.74	12.88	+115.4%	33.08	NO-CEILING-DATA	9.17/13.52 ⚐	ok
TB-01 PLATEAU	36.42	9.95	+266%	36.42	OK	10.25/10.45 ⚐	ok
BC-01 PLATEAU	31.5	14.82	+112.5%	32.07	OK	9.83/15.56 ⚐	ok
SW-01 PLATEAU	22.92	13.8	+66.1%	32.79	OK	8.17/14.49 ⚐	ok
RC-01 PLATEAU	13.32	12.23	+8.9%	22.96	OK	5.42/12.84 ⚐	ok
AP-01 PLATEAU	7.84	10.6	-26.1%	26.4	OK	1.75/11.13	under-cure
IL-01 MARCHING	23.39	27.51	-15%	49.5	NO-CEILING-DATA	8.42/28.89	under-cure
UB-01 PLATEAU	35.68	12.71	+180.7%	36.6	OK	10.5/13.35 ⚐	ok

ESTIMATED LAYERS — BY CONSTRUCTION (INTERPOLATED BETWEEN BRACKETING TCS; NOT IN THE HEADLINE)

SPOT	COMPOUND · LAYER	TMAX	CEQ@PO	CEQ@DEF	%CEF	VERDICT
BSK-T	SK-01 · Belt skim (above the belts, under the tread base) — top side	173.6	13.57	14.28	114.6	PASS
BSK-B	SK-01 · Belt skim (above the belts, under the tread base) — bottom side	173.5	13.47	14.17	113.7	PASS
INS-T	IN-01 · Insulation (between inner liner and body ply) — top side	174.5	25.59	26.32	203	PASS
INS-B	IN-01 · Insulation (between inner liner and body ply) — bottom side	167.9	16.29	16.79	129.2	PASS
BWC-T	BW-01 · Bead wire coat (bead core, under the apex) — top side	168.7	10	10.59	39.8	PASS
BWC-B	BW-01 · Bead wire coat (bead core, under the apex) — bottom side	152.2	2.78	2.97	11.1	FAIL rev. 3%

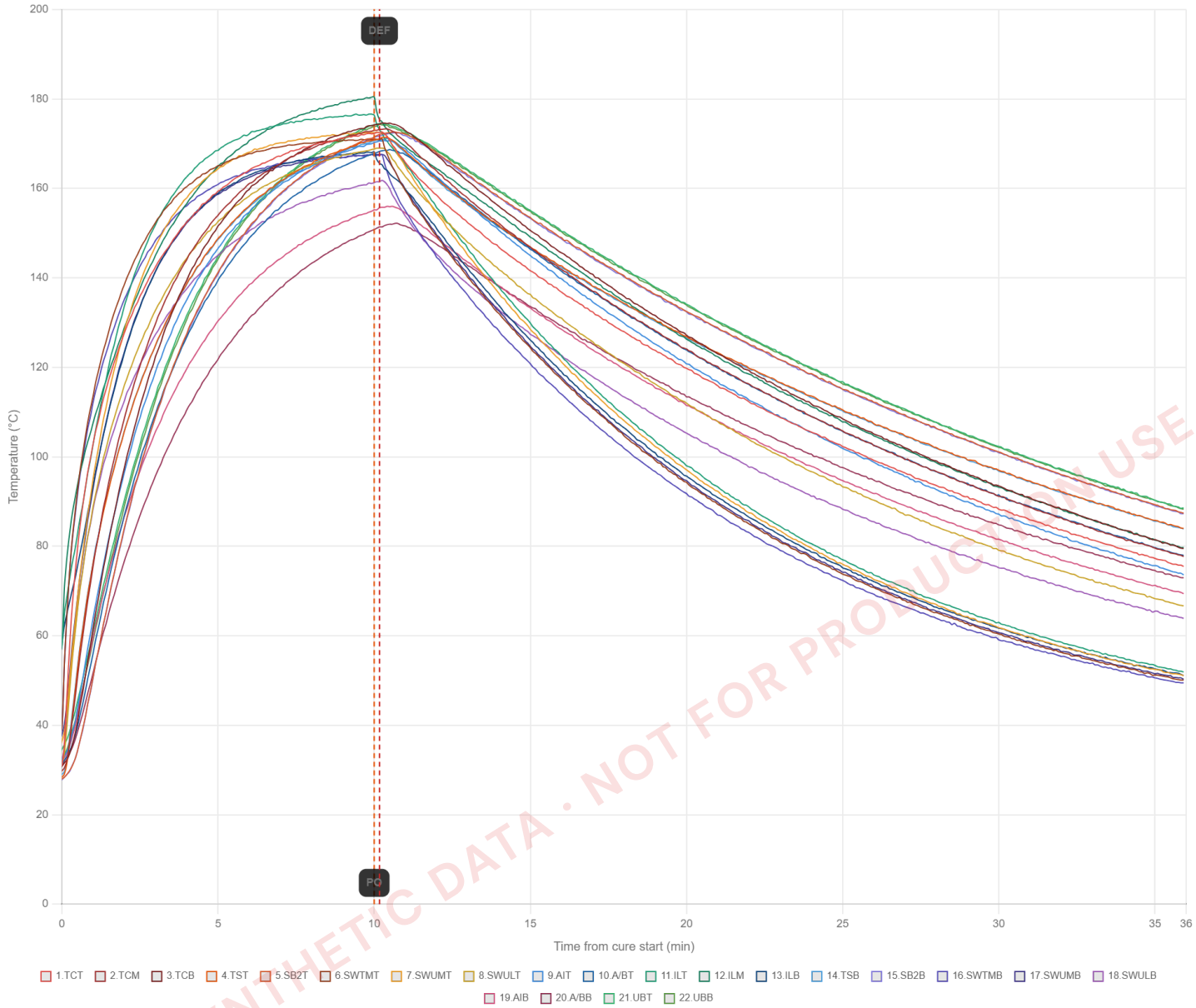
DEMO · SYNTHETIC DATA · NOT FOR PRODUCTION USE

POST-CURE ACCUMULATION (AFTER PRESS-OPEN)

CH	POSITION	CEQ@PO	CEQ@END	POST-CURE %	%CEF@PO	%CEF@END
1	TCT TR-TOUR	17.86	28.19	+58%	138.1	218.1
2	TCM TR-TOUR	15.66	29.12	+86%	121.1	225.2
3	TCB TB-01	15.7	32.11	+105%	155.1	317.3
4	TST TR-TOUR	12.65	25.58	+102%	97.9	197.9
5	SB2T BC-01	12.21	31.16	+155%	83.9	214.2
6	SWTMT SW-01	21.79	27.49	+26%	156.6	197.6
7	SWUMT SW-01	21.77	29.29	+35%	156.5	210.6
8	SWULT RC-01	13.63	21.58	+58%	110	174
9	AIT AP-01	12.43	24.76	+99%	116.1	231.1
10	A/BT AP-01	9.48	21.83	+130%	88.5	203.8
11	ILT IL-01	33.66	42.13	+25%	122	152.6
12	ILM IL-01	34.04	49.96	+47%	123.3	181
13	ILB IL-01	18.06	23.75	+32%	65.4	86.1
14	TSB TR-TOUR	12.45	25.18	+102%	96.3	194.8
15	SB2B BC-01	12	30.62	+155%	82.5	210.5
16	SWTMB SW-01	17.23	21.59	+25%	123.9	155.2
17	SWUMB SW-01	15.71	21.29	+36%	113	153.1
18	SWULB RC-01	8.42	12.92	+53%	67.9	104.2
19	AIB AP-01	4.38	9.53	+118%	40.9	89
20	A/BB AP-01	2.92	7.66	+162%	27.3	71.5
21	UBT UB-01	13.34	32.51	+144%	103	251
22	UBB UB-01	12.99	31.79	+145%	100.3	245.5

DEMO · SYNTHETIC DATA · NOT FOR PRODUCTION USE

TEMPERATURE PROFILE



DEMO · SYNTHETIC DATA · NOT FOR PRODUCTION USE

SIGN-OFF

Section cut performed: _____ All TC locations verified: _____

Tester: _____ Approved by: _____

Comments: _____